

Data Communication & Computer Networks

Course Instructor:

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Today's Lecture

What *is* the Internet?

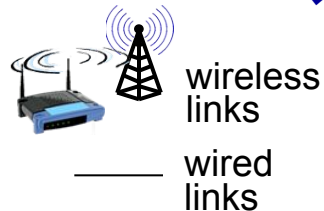
Network edge

- end systems, access networks, links

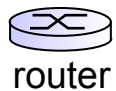
What's the Internet: “nuts and bolts” view



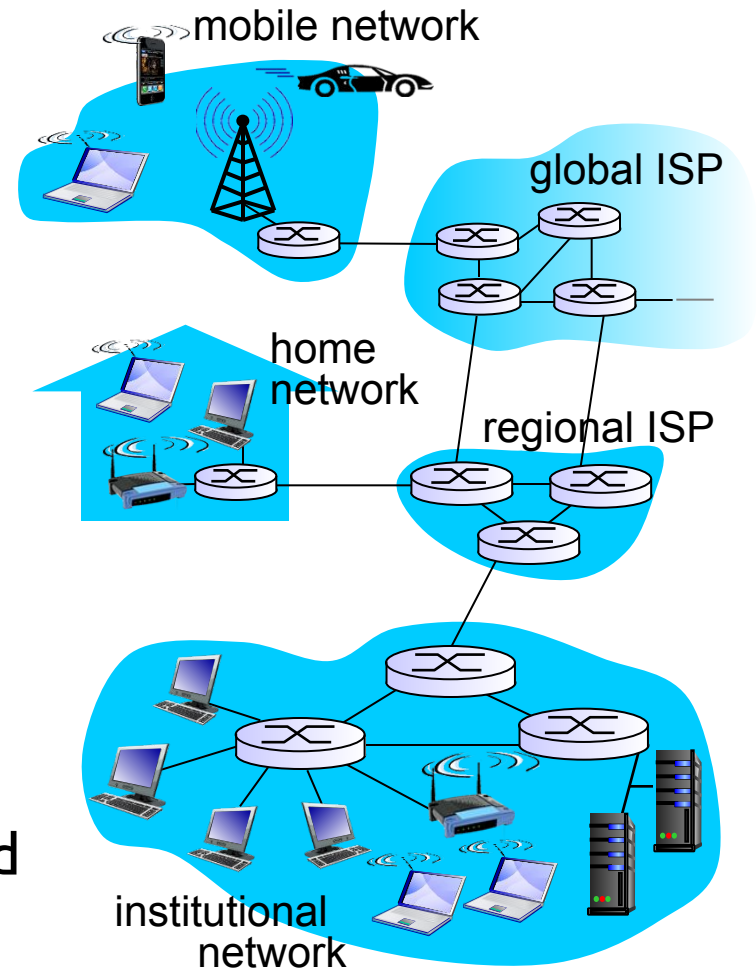
- ❖ millions of connected computing devices:
 - *hosts* = *end systems*
 - running *network apps*



- ❖ *communication links*
 - fiber, copper, radio, satellite
 - transmission rate: *bandwidth*

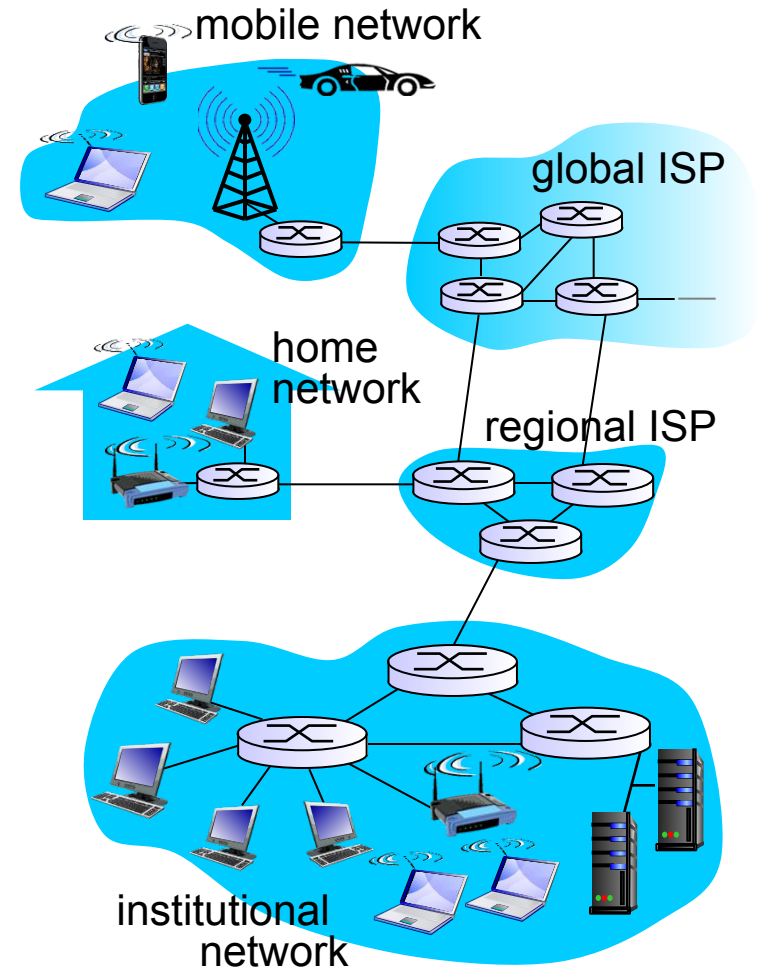


- ❖ *Packet switches*: forward packets (chunks of data)
 - *routers* and *switches*



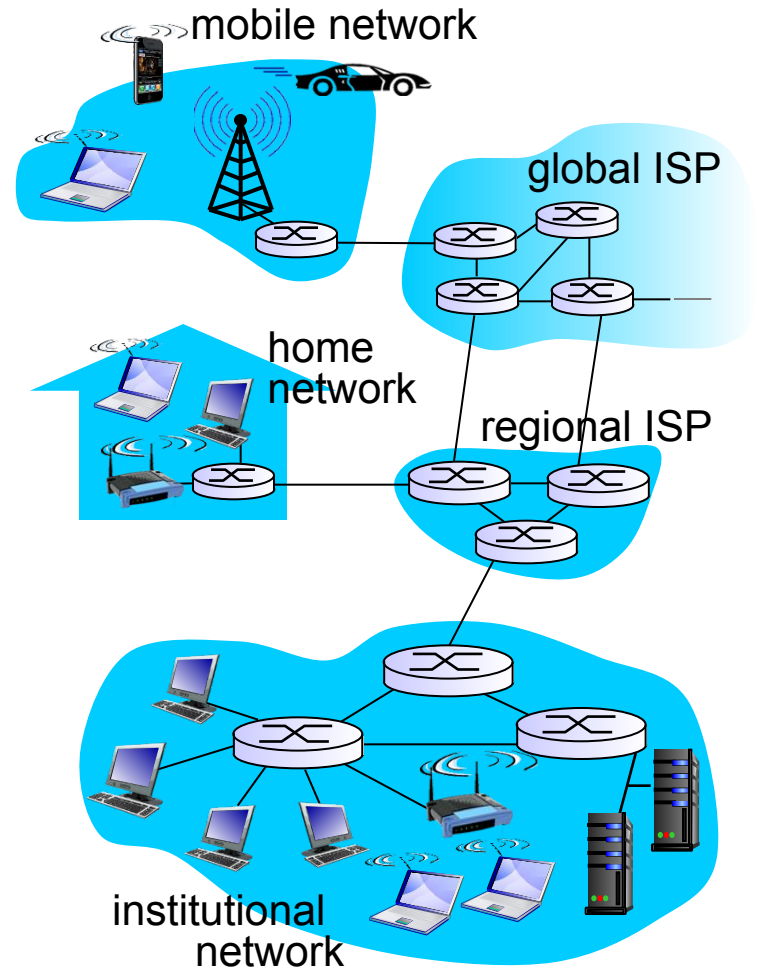
What's the Internet: “nuts and bolts” view

- ❖ *Internet: “network of networks”*
 - Interconnected ISPs
- ❖ *protocols* control sending, receiving of msgs
 - e.g., TCP, IP, HTTP, Skype, 802.11
- ❖ *Internet standards*
 - RFC: Request for comments
 - IETF: Internet Engineering Task Force



What's the Internet: a service view

- ❖ *Infrastructure that provides services to applications:*
 - Web, VoIP, email, games, e-commerce, social nets, ...
- ❖ *provides programming interface to apps*
 - hooks that allow sending and receiving app programs to “connect” to Internet
 - provides service options



What's a protocol?

human protocols:

- ❖ “what’s the time?”
 - ❖ “I have a question”
 - ❖ introductions
- ... specific msgs sent
- ... specific actions taken
when msgs received, or
other events

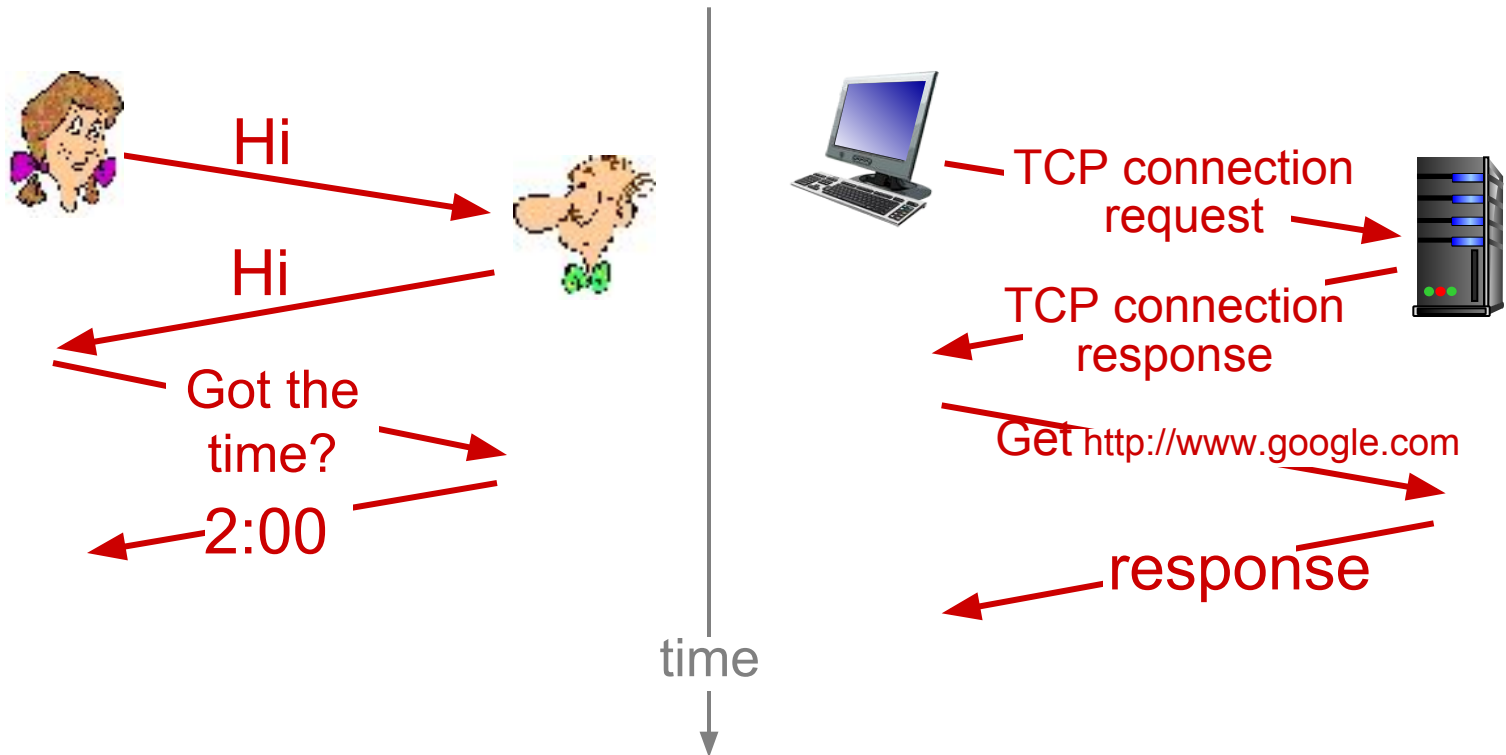
network protocols:

- ❖ machines rather than humans
- ❖ all communication activity in Internet governed by protocols

*protocols define format,
order of msgs sent and
received among network
entities, and actions taken
on msg transmission,
receipt*

What's a protocol?

a human protocol and a computer network protocol:



Roadmap

What *is* the Internet?

Network edge

- end systems, access networks, links

A closer look at network structure:

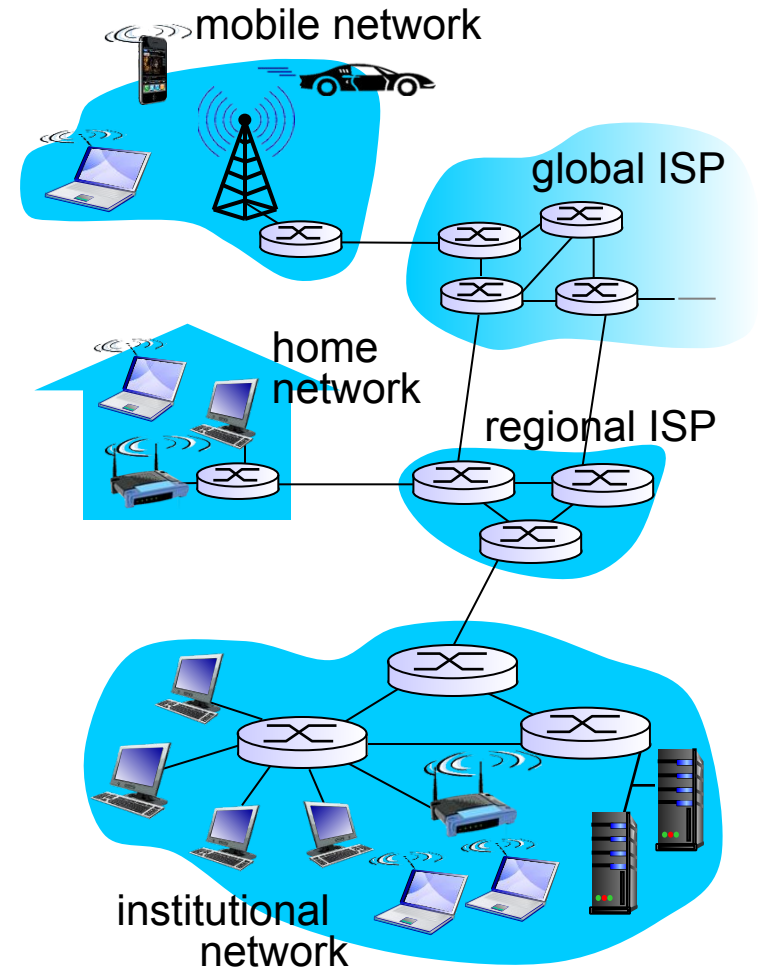
❖ *network edge:*

- hosts: clients and servers
- servers often in data centers

❖ *access networks, physical media:* wired, wireless communication links

❖ *network core:*

- interconnected routers
- network of networks



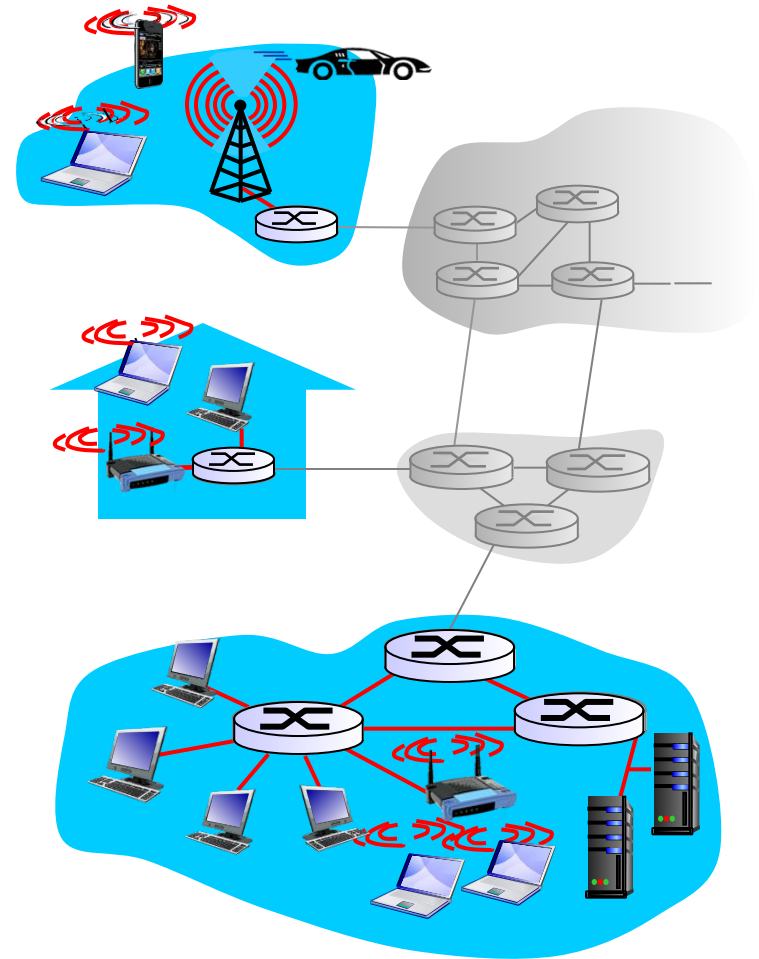
Access networks and physical media

Q: How to connect end systems to edge router?

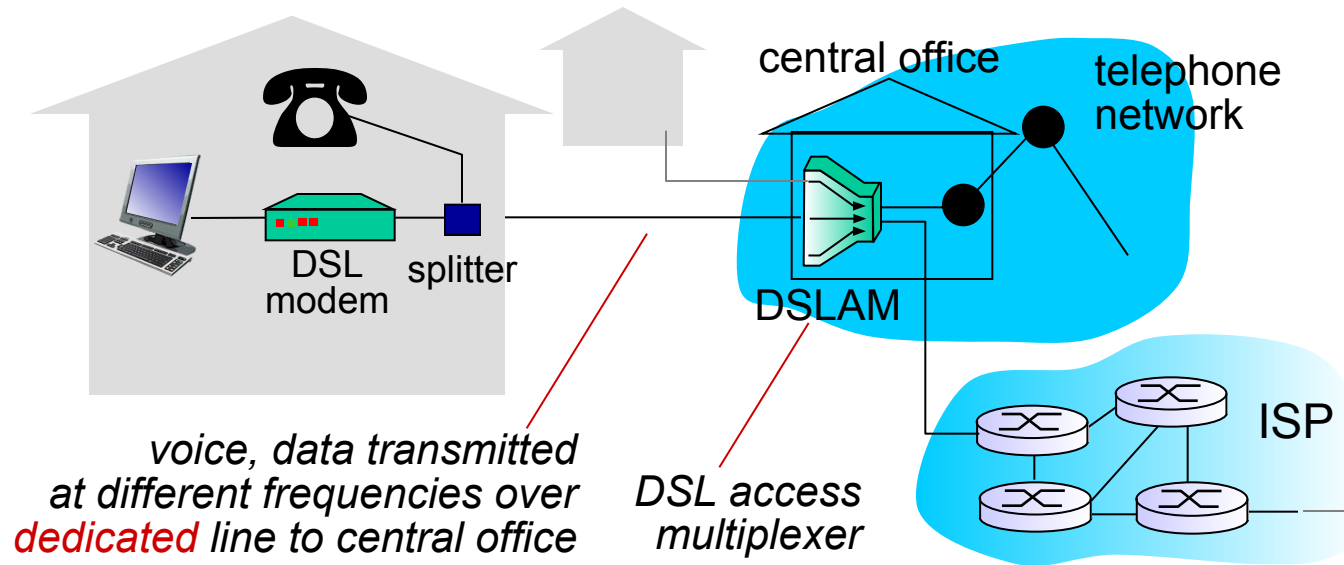
- ❖ residential access nets
- ❖ institutional access networks (school, company)
- ❖ mobile access networks

keep in mind:

- ❖ bandwidth (bits per second) of access network?
- ❖ shared or dedicated?

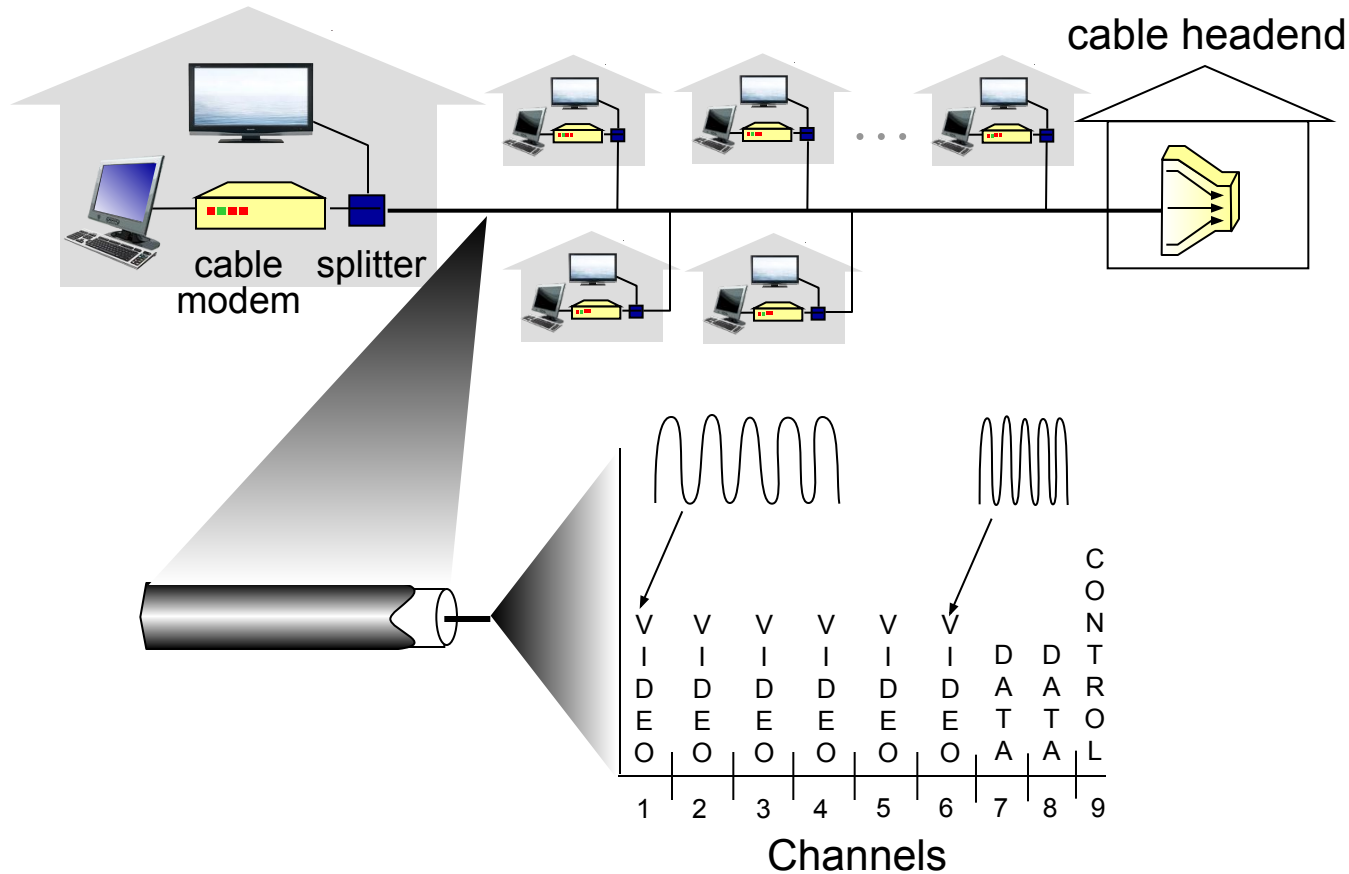


Access net: digital subscriber line (DSL)



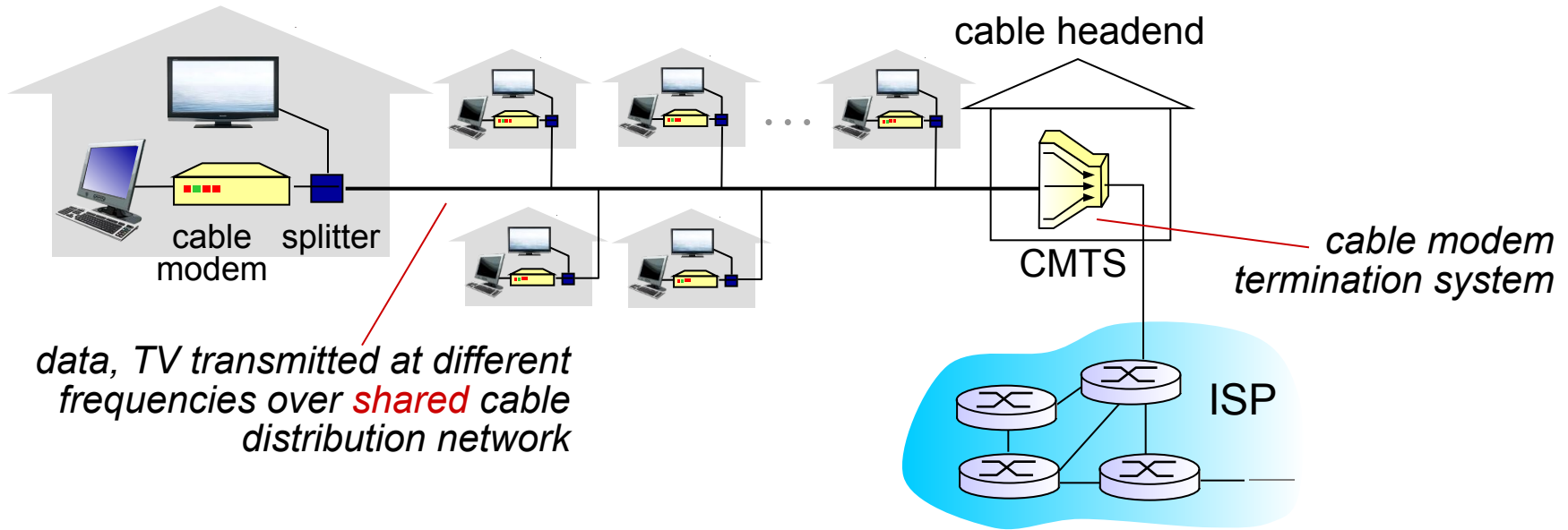
- ❖ use *existing* telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- ❖ < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
- ❖ < 24 Mbps downstream transmission rate (typically < 10 Mbps)

Access net: cable network



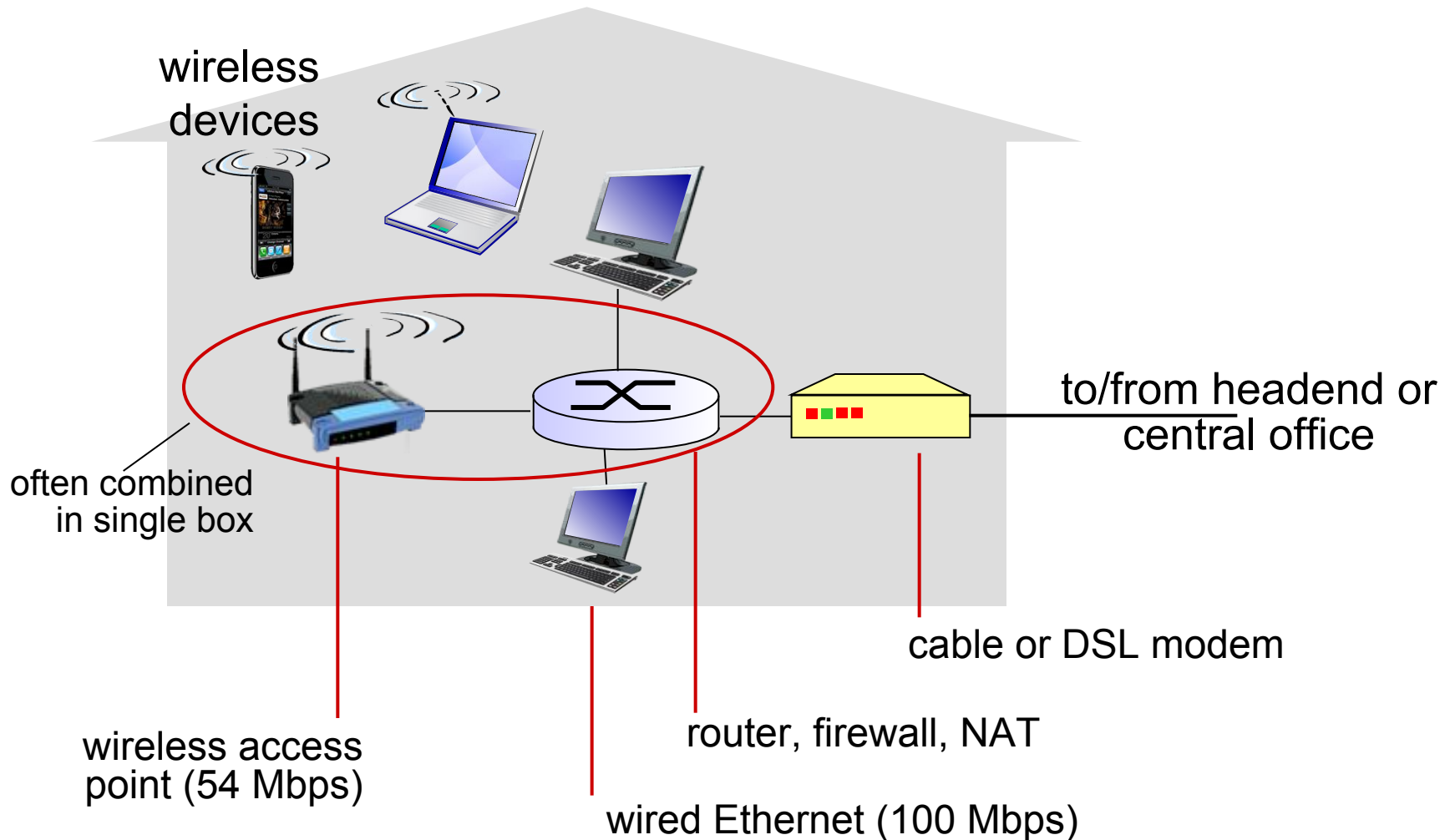
frequency division multiplexing: different channels transmitted in different frequency bands

Access net: cable network

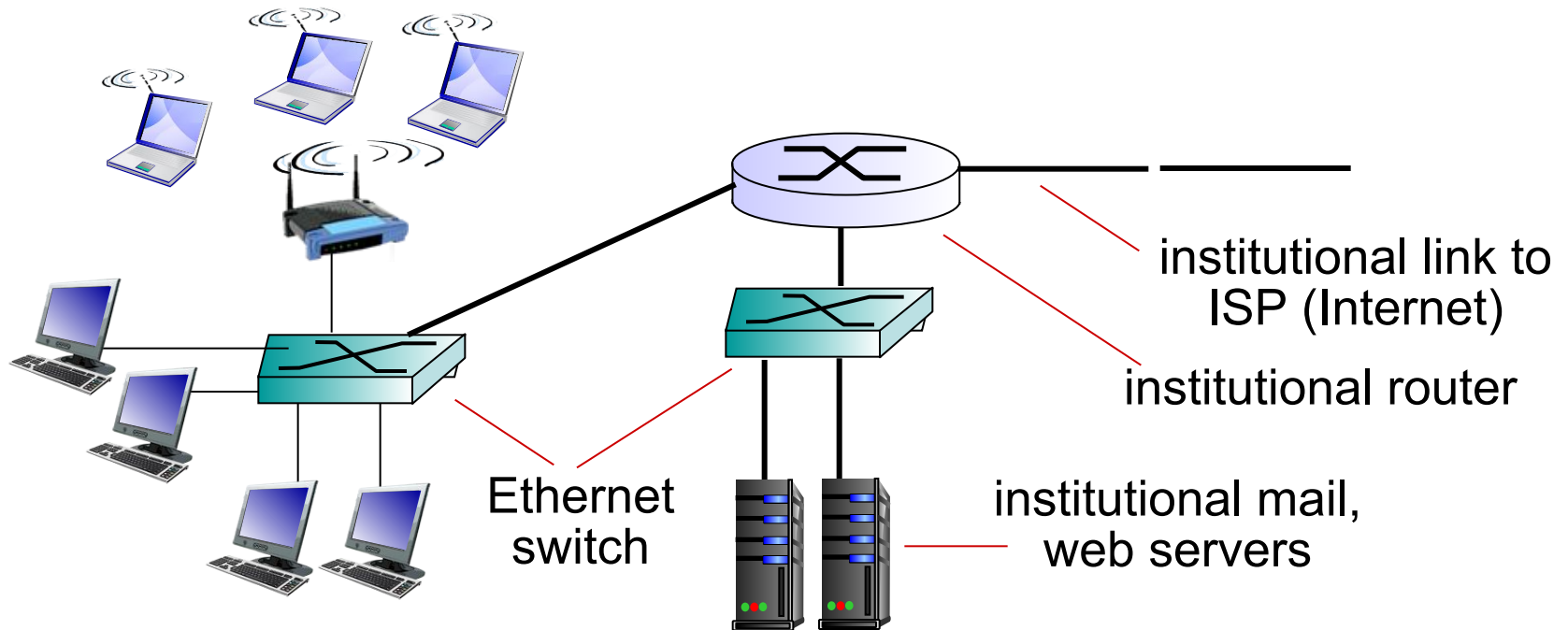


- ❖ **HFC: hybrid fiber coax**
 - up to 30Mbps downstream transmission rate, 2 Mbps upstream transmission rate
- ❖ **network** of cable, fiber attaches homes to ISP router
 - homes *share access network* to cable headend

Access net: home network



Enterprise access networks (Ethernet)



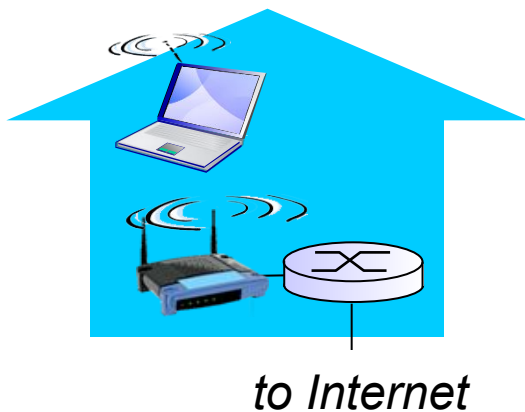
- ❖ typically used in companies, universities, etc
- ❖ 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- ❖ today, end systems typically connect into Ethernet switch

Wireless access networks

- ❖ shared *wireless* access network connects end system to router
 - via base station aka “access point”

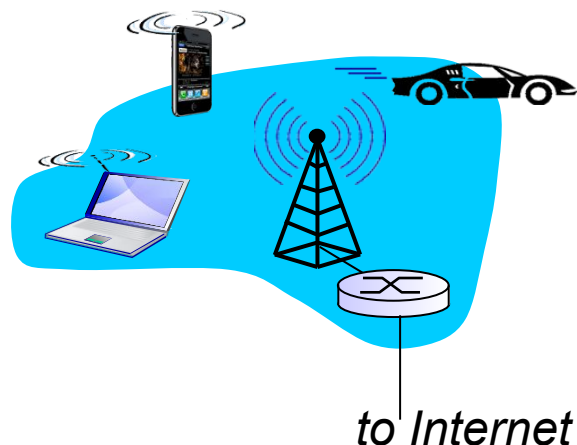
wireless LANs:

- within building (100 ft)
- 802.11b/g (WiFi): 11, 54 Mbps transmission rate



wide-area wireless access

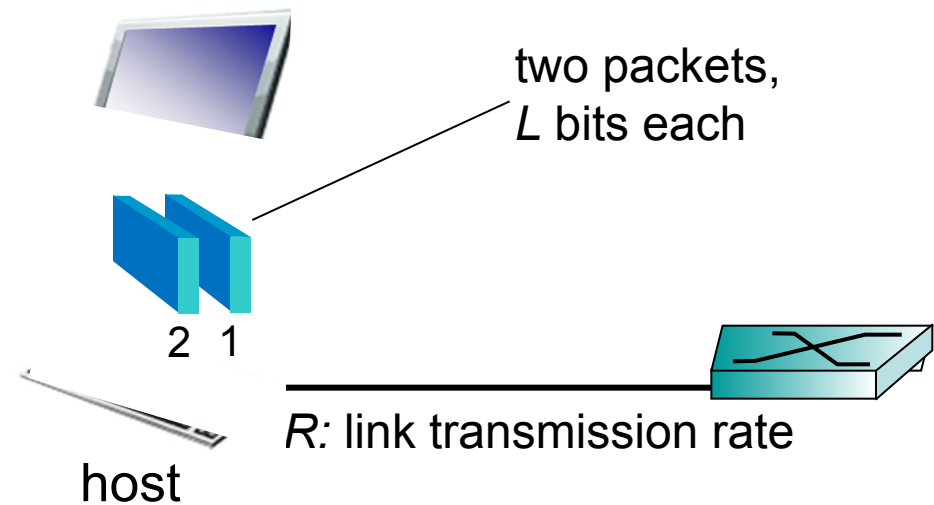
- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE



Host: sends *packets* of data

host sending function:

- ❖ takes application message
- ❖ breaks into smaller chunks, known as *packets*, of length L bits
- ❖ transmits packet into access network at *transmission rate* R
 - link transmission rate, aka link *capacity*, aka link *bandwidth*



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Physical media

- ❖ **bit:** propagates between transmitter/receiver pairs
- ❖ **physical link:** what lies between transmitter & receiver
- ❖ **guided media:**
 - signals propagate in solid media: copper, fiber, coax
- ❖ **unguided media:**
 - signals propagate freely, e.g., radio

twisted pair (TP)

- ❖ two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps



Physical media: coax, fiber

coaxial cable:

- ❖ two concentric copper conductors
- ❖ bidirectional
- ❖ broadband:
 - multiple channels on cable
 - HFC



fiber optic cable:

- ❖ glass fiber carrying light pulses, each pulse a bit
- ❖ high-speed operation:
 - high-speed point-to-point transmission (e.g., 10's-100's Gpbs transmission rate)
- ❖ low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Physical media: radio

- ❖ signal carried in electromagnetic spectrum
- ❖ no physical “wire”
- ❖ bidirectional
- ❖ propagation environment effects:
 - reflection
 - obstruction by objects
 - interference

radio link types:

- ❖ **terrestrial microwave**
 - e.g. up to 45 Mbps channels
- ❖ **LAN** (e.g., WiFi)
 - 11 Mbps, 54 Mbps
- ❖ **wide-area** (e.g., cellular)
 - 3G cellular: ~ few Mbps
- ❖ **satellite**
 - Kbps to 45Mbps channel (or multiple smaller channels)
 - 270 msec end-end delay

- ❖ Lets take an example of a letter
- ❖ They do. Through what? LETTERS. Right?
- ❖ So What was the procedure of sending letters?
- ❖ Few steps
- ❖ Lets write them down

1. **Paper/pencil** was required
2. Some **language/codes** were used
3. **Post Offices** helps in sending those letters
4. **Stamps** were used to ensure the package weight
5. But not without **City Address**
6. **Home and Street addresses** were required so the letter can be posted to the right person.
7. Finally your letter is mailed via some **route physically** that is decided by the post office

Application (7) SMTP, FTP, Telnet

Presentation (6) Format Data, Encryption

Session (5) Start & Stop Sessions

Transport (4) TCP, UDP, Port Numbers

Network (3) IP Address, Routers

Data Link (2) MAC Address, Switches

Physical (1) Cable, Network Interface Cards, Hubs

OSI Model

- ❖ **Application** is same as the paper/pencil requirement. If you need to send some mail you choose the SMTP, if some file is required to send FTP application is used and if simple bidirectional text oriented communication is occurring Telnet application will be used.
- ❖ **Presentation** we all know the computer understands the binary language and transforms data into signal before sending. So the encryption/decryption(coding/decoding) task is done here

OSI Model (continued)

- ❖ What if post offices aren't available. We couldn't be able to initialize our letter/communication. So **sessions** are to alert **servers** that some application of required server is needed to be used and start session.
- ❖ Stamps idea that contains that fee is dropped because networking system allowed us to do that task freely and replaced with **Transport** that decides whether the connection will be connectionless(UDP) or connection oriented (TCP/IP)

OSI Model (continued)

- ❖ **Network layer** manages and checks for the IP address
- ❖ While **Data Link** Layer decides the physical address of device known as MAC Address just as the home address
- ❖ When everything gets ensured all the required data is available and finally the mail is posted or the said data is sent **physically** via cable/signals

Protocol “layers”

*Networks are
complex,
with many “pieces”:*

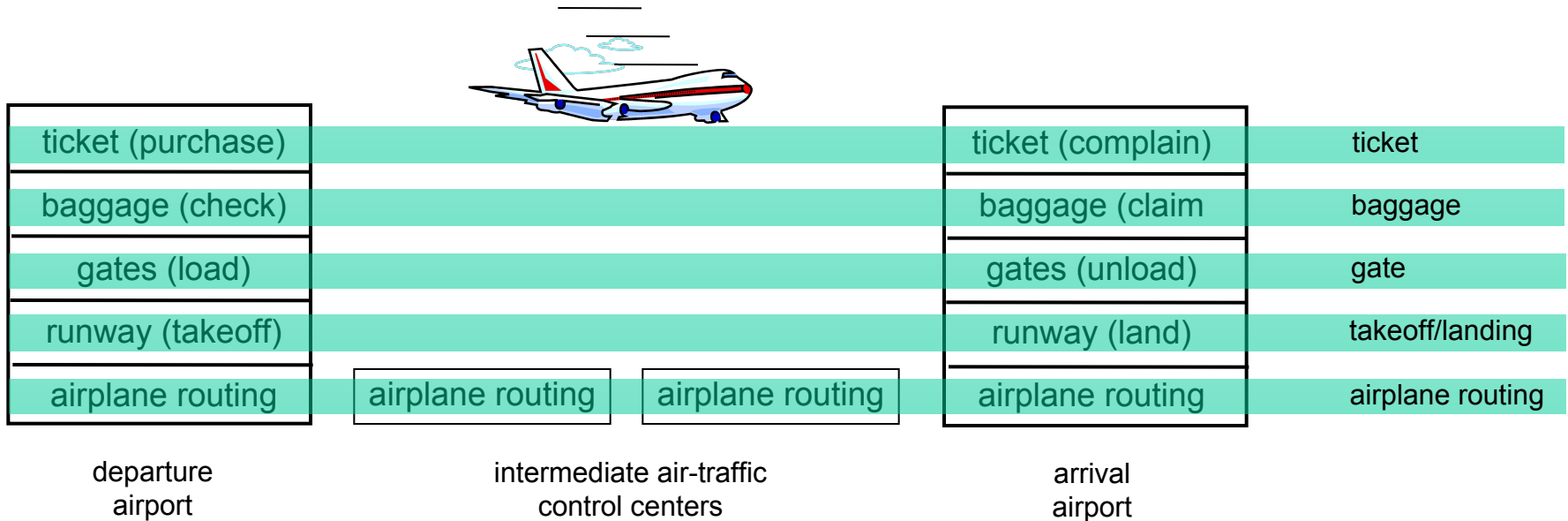
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Organization of air travel



❖ a series of steps

Layering of airline functionality



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

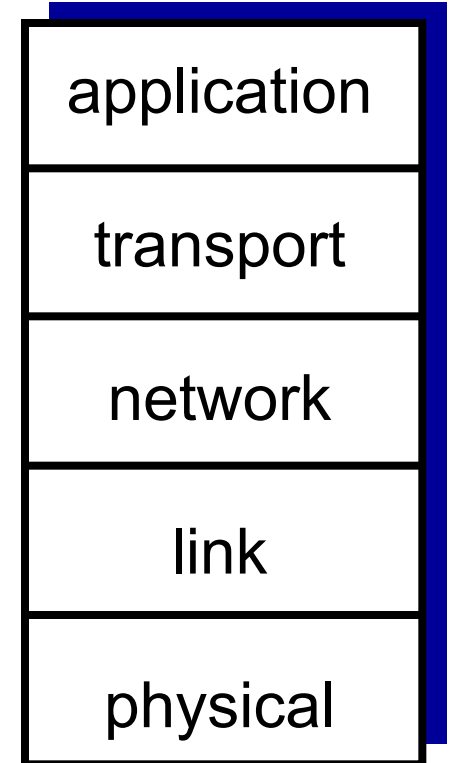
Why layering?

dealing with complex systems:

- ❖ allows identification, relationship of complex system's pieces
 - layered *reference model* for discussion
- ❖ Layers eases maintenance, updating of system
 - change of implementation of layer's service transparent to rest of system
 - e.g., change in port procedure doesn't affect rest of system
- ❖ layering considered harmful?

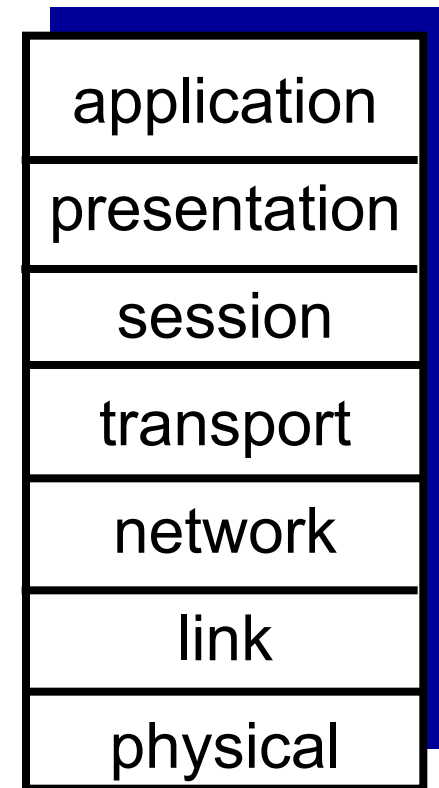
Internet protocol stack

- ❖ *application*: supporting network applications
 - FTP, SMTP, HTTP
- ❖ *transport*: process-process data transfer
 - TCP, UDP
- ❖ *network*: routing of datagrams from source to destination
 - IP, routing protocols
- ❖ *link*: data transfer between neighboring network elements
 - Ethernet, 802.111 (WiFi), PPP
(Point-to-Point Protocol)
- ❖ *physical*: bits “on the wire”

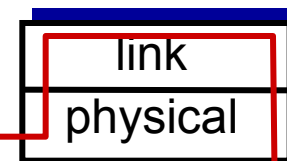
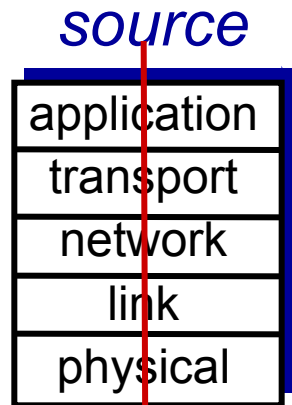
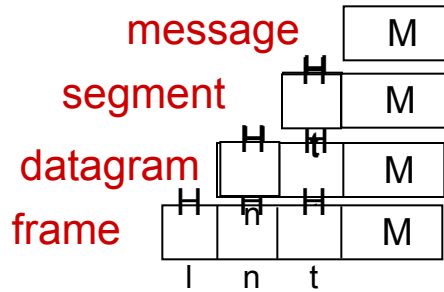


ISO/OSI reference model

- ❖ *presentation*: allow applications to interpret meaning of data, e.g., encryption, compression, *session*: synchronization, checkpointing, recovery of data exchange
- ❖ Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in application
 - needed?

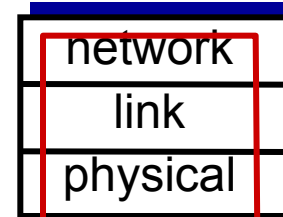
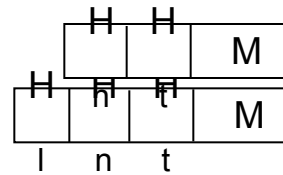
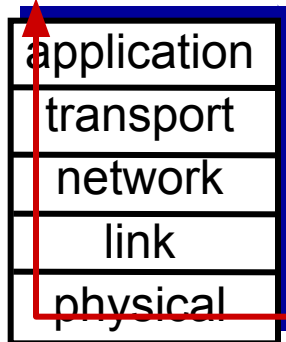


Encapsulation



switch

destination



router